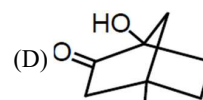
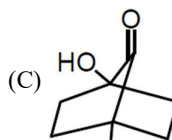
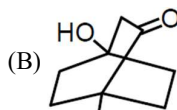
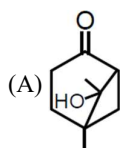
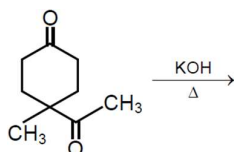
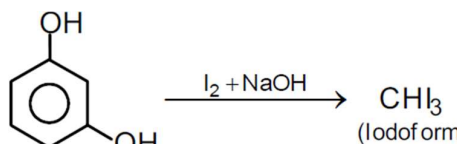
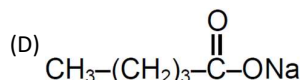
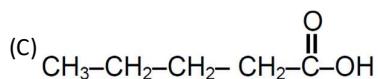
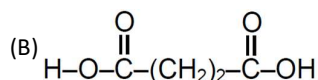
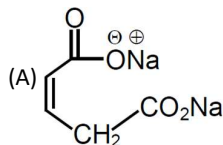


SCQ (Single Correct Type) :

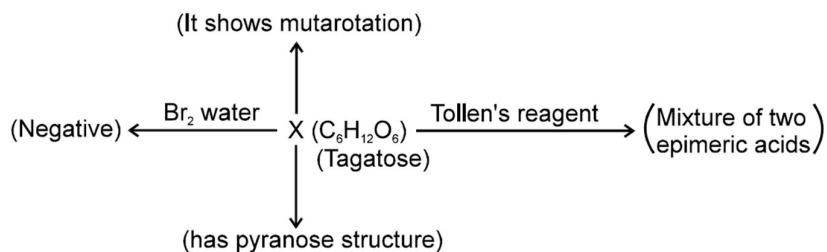
1. The major product of the following reaction is –

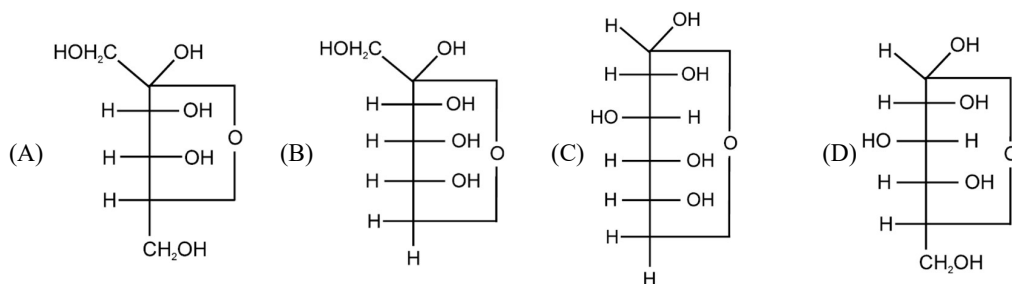


2.  CHI_3 + Compound (P) is.

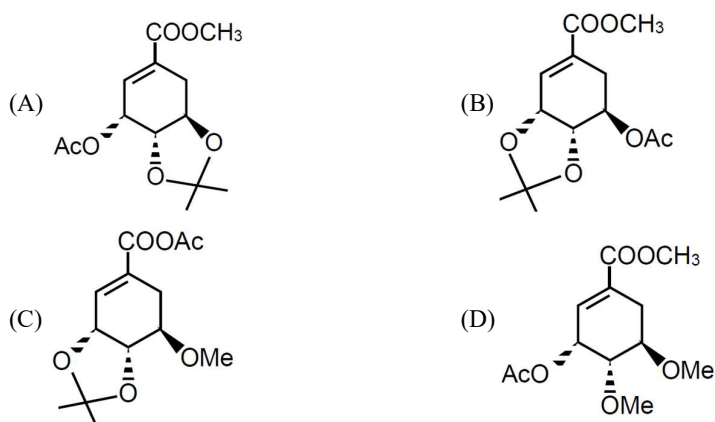
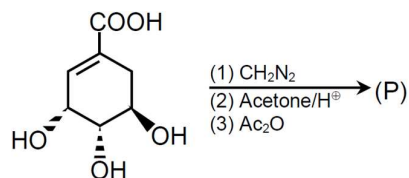


3. Observe the following road map and identify the carbohydrate 'X' (Tagatose)



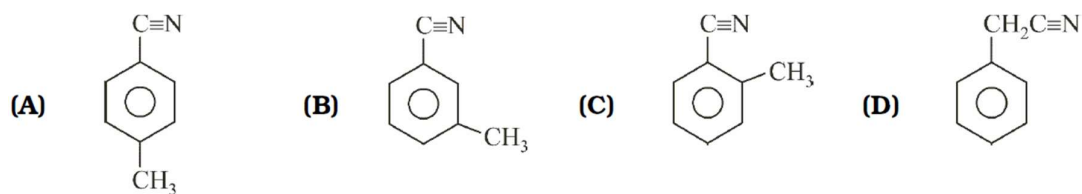


4. (P) decolourise $\text{Br}_2 / \text{H}_2\text{O}$. Identify the structure of (P).



Compound (X) ($\text{C}_8 \text{H}_7 \text{N}$) on treatment with cold $\text{SnCl}_2 / \text{dil. HCl}$ (aqueous), produces $\text{Y} (\text{C}_8 \text{H}_8 \text{O})$. (Y) on heating with sodium acetate in acetic anhydride followed by acidification forms (Z) ($\text{C}_{10} \text{H}_{10} \text{O}_2$). All these compounds (X,Y,Z) on oxidation under severe conditions form a diacid which gives only one mononitro isomer in nitrating mixture.

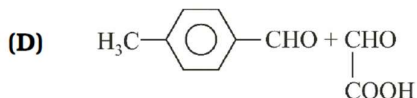
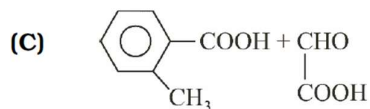
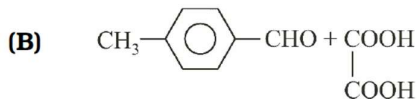
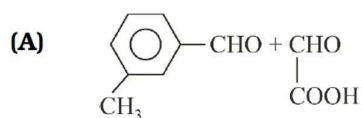
5. What is structure of (X) :



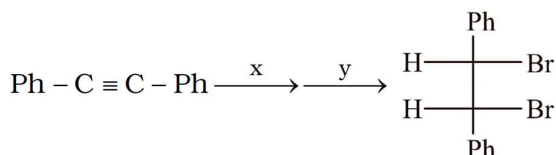
6. What is not true about (Y) :

- (A) It cannot give Fehling's solution test (B) It gives Iodoform test positive
(C) It gives Tollen's test positive (D) it gives 2, 4-DNP test positive

7. The reductive ozonolysis products of (Z) are :



8. The reagents x and y are respectively:



(A) $\text{Br}_3 / \text{CCl}_4$, H_2 / Ni

(B) HBr , HBr

(C) $\text{Na}/\text{NH}_3(\ell)$, $\text{Br}_2 / \text{CCl}_4$

(D) $\text{Br}_2 / \text{CCl}_4$, Lindlar's hydrogenation

9. Statement-1: 2-Bromobutane on heating with alcoholic KOH gives 3 isomeric alkenes which on treatment with $\text{Br}_2 / \text{CCl}_4$ give 5 products.

Statement-2: Stereochemical nature of elimination by alcoholic KOH and addition of Br_2 is ANTI and SYN respectively.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement -2 is False

(D) Statement-1 is False, Statement-2 is True

10. Statement-1: Hydrolysis of sucrose brings a change in sign of rotation towards plane polarized light.

Statement-2: Fructose has specific rotation -92.4° and glucose has $+52.5^\circ$.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

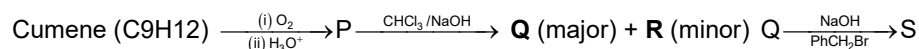
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement -2 is False

(D) Statement-1 is False, Statement-2 is True

MCQ (Multiple Correct Type) :

11. The correct statement(s) about the following reaction sequence is(are)



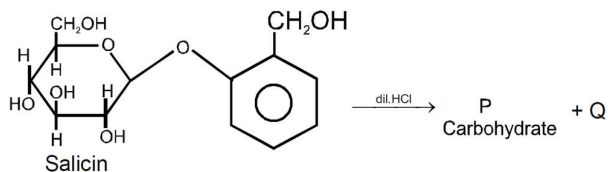
(A) **R** is steam volatile

(B) **Q** gives dark violet coloration with 1% aqueous FeCl_3 solution

(C) **S** gives yellow precipitate with 2, 4-dinitrophenylhydrazine

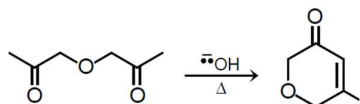
(D) **S** gives dark violet coloration with 1% aqueous FeCl_3 solution

12. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain. Observe the following reaction of salicin

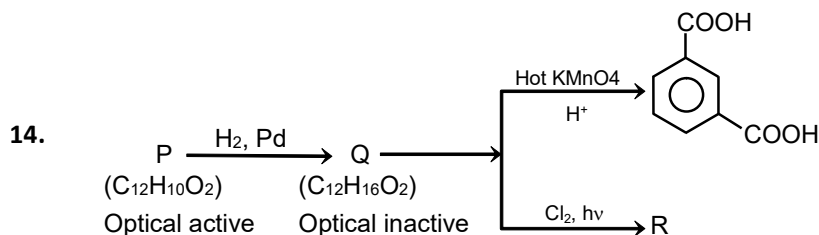


The correct statement (s) is (are)

- (A) P is D- glucose (B) Q is 2-hydroxy benzylalcohol
(C) Q is 2-Methoxy benzylalcohol (D) The above reaction occurs through a carbocation
13. Which of the following step, mechanism/s are involved in above conversion.



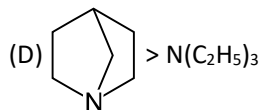
- (A) Acid base (B) Nucleophilic addition
(C) E1cB (D) Dehydration



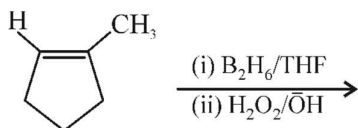
P gives CO_2 gas with NaHCO_3 (No sp hybridised carbon present in compound 'P')

Correct statement(s) is(are) for the above reactions :

- (A) Total monochloro products of compound (Q) including stereoisomers is 7.
(B) IUPAC name of (P) is 3-(1-Ethynylprop-2-enyl)benzene carboxylic acid.
(C) Compound (Q) gives effervescence of CO_2 with sodium bicarbonate.
(D) One monochloro product of Q is achiral.
15. The correct order of nucleophilicity for following species is/are :
- (A) $\text{NH}_2^- > \text{NH}_3 > \text{Ph}-\ddot{\text{N}}\text{H}_2$ (B) $\text{Me}-\text{O}^- > \text{Me}_2\text{CH}-\text{O}^- > \text{Me}_3\text{C}-\text{O}^-$
(C) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ (in polar protic solvent)



16.



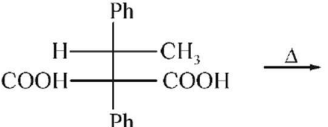
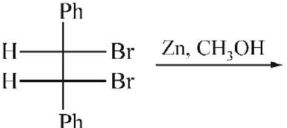
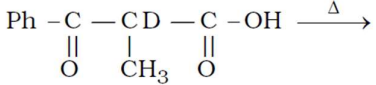
True statement about above reaction

- (A) Reagent involve stereospecific SYN addition of H and OH species
(B) Product obtained is trans isomer
(C) BH_3 act as electrophile
(D) Two stereoisomers are obtained as product

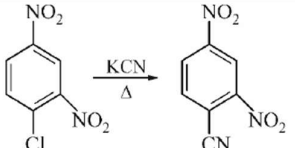
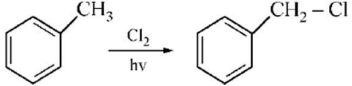
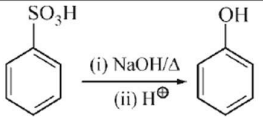
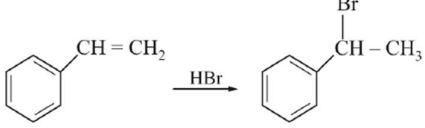
17. Select the correct statements out of the following:

- (A) β – D(+) glucopyranose is more stable than α – D(+) glucopyranose
 (B) sucrose is laevorotatory sugar
 (C) Although fructose is a ketone but it gives positive Tollen's test
 (D) Starch contains a- glycosidic linkage whereas cellulose contains b- glycosidic linkage

18. Match the reactions given in column I with their properties listed in column II.

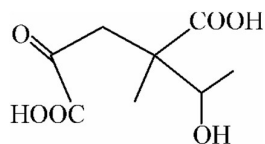
Column-I		Column-II	
(A)		(p)	Anti elimination
(B)	$\text{Ph} - \text{C} \equiv \text{C} - \text{Ph} \xrightarrow[\text{poisoned}]{\text{H}_2, \text{Pd} - \text{BaSO}_4}$	(q)	Stereoselective
(C)		(r)	Optical diastereomers
(D)		(s)	CO ₂ will evolve
		(t)	Racemic mixture

19.

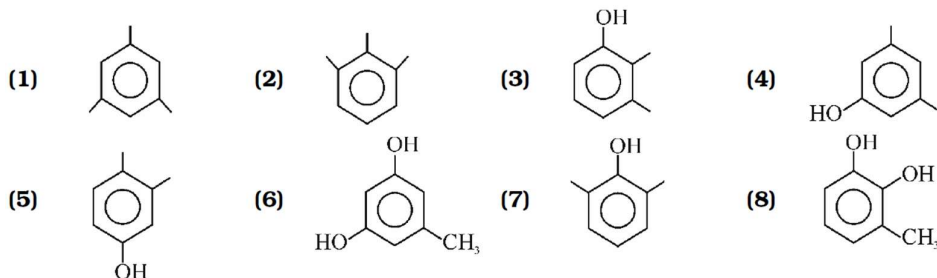
Column-I		Column-II	
(A)		(p)	Substitution reaction
(B)		(q)	Free radical reaction
(C)		(r)	Nucleophile is the attacking species
(D)		(s)	Electrophile is the attacking species
		(t)	Addition reaction

(Numerical Value type) :

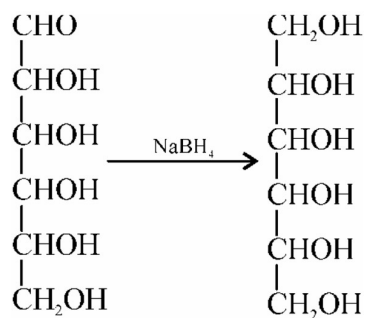
20. An optically active compound (M) on ozonolysis gives following structure in absence of zinc.



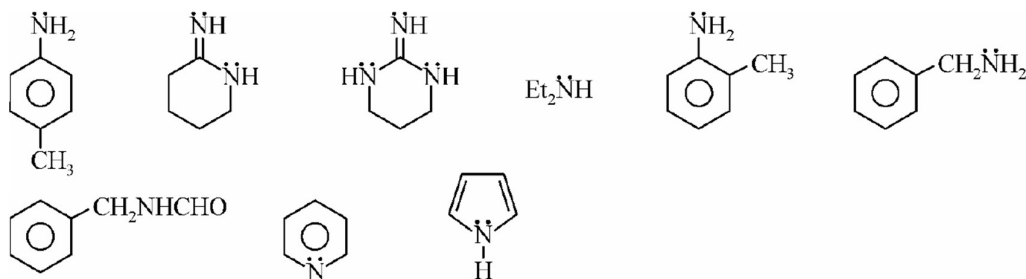
If (M) does not respond to NaHCO_3 test, what product is obtained when (M) is heated in presence of conc. H_2SO_4 as catalyst. Fill the number against product in key :



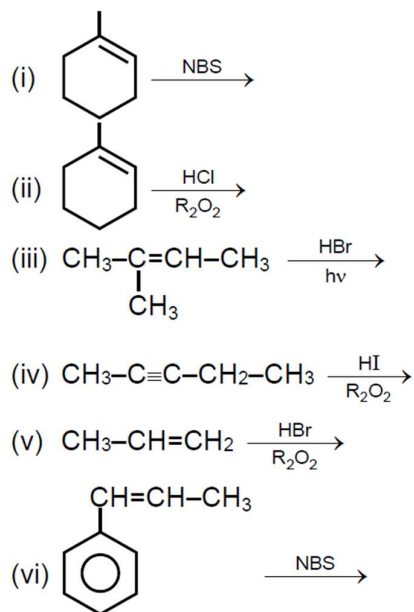
21. Observe the following reaction and find out that how many number of reactant stereoisomers can be reduced to optically inactive meso products.



22. How many of the following are stronger bases than aniline ?



23. How many reactions will proceed through free radical addition mechanism ?



Answer Key

1. (B)	2. (A)	3. (B)	4. 5.	(A)	6. (B)	7. (D)
8. (C)	9. (C)	10. (A)	11. (B, C)	12. (A,B,C,D)		
13. (ABCD)	14. (ABCD)	15. (A,B, D)	16. ABCD			
17. ACD	18. A – rs ; B-q; C-pq ; D-st	19. A-pr ; B – pq ; C-pr ; D – st				
20. 5	21. 4	22. 6	23. 02.00			